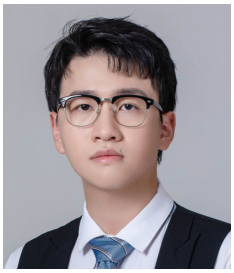
Wang Yao (王曜)



Postdoctoral Research Fellow

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Biography

Yao Wang is a Postdoctoral Research Fellow at Nanyang Technological University, Singapore (NTU), working at the intersection of **waste management, climate change, and One Health**. He investigates solid waste biogeochemistry, pollutant release, and greenhouse gas emission mechanisms to inform sustainable waste management. He has published **18 papers in leading journals**, including sole-first-author articles in ***Nature Sustainability*** and ***One Earth***, and co-authored articles in ***Nature Climate Change***, ***Nature Reviews Materials***, and ***Nature Reviews Earth & Environment***.

His research has been adopted by **Climate TRACE** [link], the world's largest independent GHG emissions inventory. He also contributed to the development of **WasteMAP** [link] with **RMI**, a global decision-support platform supporting waste methane assessment and mitigation planning in countries including the Philippines, Mexico, Brazil, and Nigeria. His findings also inform policy assessments by the **OECD** and the **U.S. Department of Energy**.

His sole-first-author *Nature Sustainability* paper was one of only 25 studies selected worldwide for the **2026 Frontiers Planet Prize**: Singapore's only selected study and the only one focused on solid waste globally. He also received the **Chinese Government Award for Outstanding Self-Financed Students Abroad**, a highly selective China Scholarship Council honour with approximately **650 recipients worldwide**.

Professional experience

Nanyang Technological University, Singapore, Postdoctoral Research Fellow *Mar 2026–present*

- **Host:** Associate Prof. Xunchang Fei

Education

Nanyang Technological University, Singapore *Aug 2022–Jan 2026*
Ph.D. in Environmental Engineering

- **Supervisor:** Associate Prof. Xunchang Fei
- **Advisory Committee:** Prof. Lim Teik Thye, Prof. Grzegorz Lisak
- **Thesis**: The biogeochemical dynamics of major elements in municipal solid waste landfills

Nanyang Technological University, Singapore *Aug 2020–May 2022*
M.Eng. in Environmental Engineering

- **Supervisor:** Associate Prof. Xunchang Fei
- **Thesis**: The degradation processes of municipal solid waste in landfills

Tongji University, B.S. in Environmental science *Sep 2016–Jul 2020*

- **Supervisor:** Prof. Daqiang Yin, Prof. Hongtao Wang
- **Thesis**: QSAR modeling of chlorinated paraffin in aquatic ecosystems

Research interests

My research addresses sustainable solid waste management in a changing Earth system. I study the two-way interactions between waste systems and global change and turn the findings into engineering and management solutions, aiming to make waste systems measurable, predictable, and manageable. My work focuses on:

Measuring global impacts of waste facilities.
By cross-checking IPCC-based bottom-up inventories against top-down atmospheric estimates, I revealed systematic underestimation of landfill methane, especially in data-scarce countries and at informal and legacy dumpsites. I combine machine learning, satellite remote sensing, and field data to map waste

facilities and build local-to-global inventories toward a global measurement, reporting, and verification (MRV) framework.
Related publication(s): FA-1, FA-3, FA-5, CA-3, CA-6.

Understanding coupled waste processes under global change.
I identify climate-sensitive controls on waste decay, such as composition, moisture, and temperature (CMT), and translate them into site-specific models, including the CMT method for landfill methane adopted by Climate TRACE. I investigate how global change, such as warming, extreme rainfall, sea-level rise, and shifting waste composition, alters coupled biological, chemical, and physical waste processes, and develop physics-informed process models to predict gas and contaminant fluxes.
Related publication(s): FA-4, CA-2, CA-5, CA-7, CA-11.

Engineering low-carbon, climate-resilient waste systems.
I combined revised emission estimates and exposure risks to develop low-cost, local mitigation strategies, from food waste reduction and waste-to-energy to landfill gas capture, and co-developed WasteMAP with RMI. I aim to design and field-verify lasting interventions, such as adaptive gas collection, biocovers, and aeration for closed landfills, carbonation of incineration ash, and climate-resilient closure designs.
Related publication(s): FA-2, FA-6, CA-1.

Tracing the long-term fate of plastics and microplastics.
I study how land-disposed plastics age, fragment into microplastics, and are released to the environment, drawing on ten-year landfill simulation experiments. I also investigate plastiglomerates, plastic–rock composites that may form in landfills, and how plastics may alter carbon and nitrogen cycling in waste, to inform monitoring, risk assessment, and end-of-life management.
Related publication(s): FA-7, CA-4, CA-8, CA-9, CA-10.

Publications

Google Scholar citations = 521, **H-index** = 10, **Average IF per paper** = 21.5 Latest update: Oct 2026

First & corresponding author

FA-1 Hong, P., **Wang, Y.***, Fei, X.*, Guo, M.*, 2026. Overlooked methane in high-density cities: from detection to verifiable mitigation. **Resources, Conservation and Recycling** (IF=10.9), 237, 109186. https://doi.org/10.1016/j.resconrec.2026.109186

FA-2 **Yao Wang**, Zhou, C., Zhang, H., Ma, S., Lou, Z., Raskin, L., Skerlos, S., Coulon, F., Zekkos, D., Hussain, A., Yadav, V., Lisak, G., Fang, M., Yin, K., He, H., Yuan Wang, Fei, X.*, 2025. Biogeochemical dynamics of major elements in municipal solid waste landfills can induce health risks for nearly 1 billion people. **One Earth** (IF=16.1), 8(9), 101418. https://doi.org/10.1016/j.oneear.2025.101418 (Citations: **16**)

FA-3 **Wang, Y.**, Yin, K., Fang, M., Fei, X.*, 2025. Underappreciated majority of solid waste: insufficient data and unclear impacts. **The Innovation** (IF=25.7), 6(3), 100776. https://doi.org/10.1016/j.xinn.2024.100776 (Citations: **4**)

FA-4 **Wang, Y.**, Fang, M., Lou, Z., He, H., Guo, Y., Pi, X., Wang, Y.J., Yin, K.*, Fei, X.*, 2024. Methane emissions from landfills differentially underestimated worldwide. **Nature Sustainability** (IF=29.2), 7, pp.496–507. https://doi.org/10.1038/s41893-024-01307-9 (Citations: **148**, ESI Highly Cited Paper).

FA-5 **Wang, Y.**, Zhou, C., Lou, Z., Zhang, H., Hussain, A., Zhan, L., Yin, K., Fang, M., Fei, X.*, 2024. Underestimated methane emissions from solid waste disposal sites reveal missed greenhouse gas mitigation opportunities. **Engineering** (IF=10.1), 36, pp.12-15. https://doi.org/10.1016/j.eng.2023.12.011 (Citations: **29**)

FA-6 **Wang, Y.**, Ruiz-Acevedo, A., Rameez, E., Raghavan, V., Hussain, A.* and Fei, X.*, 2024. Toward sustainable waste management in small islands developing states: integrated waste-to-energy solutions in Maldives context. **Frontiers of Environmental Science & Engineering** (IF=6.7), 18(2), 24. https://doi.org/10.1007/s11783-024-1784-7 (Citations: **22**)

FA-7 **Wang, Y.**, Lu, X. and Fei, X.*, 2021. Property changes of conventional plastic waste mixed with municipal solid waste after 10-year degradation experiments simulating landfill conditions. **Journal of Hazardous Materials Letters** (IF=6.6), 2, p.100047. https://doi.org/10.1016/j.hazl.2021.100047 (Citations: **26**)

Contributing author

CA-1 Yin, K.*, Zhu, J., Wu, M., Zhang, H., Ling, X., Cai, M., Ren, S., Zhao, X., Ling, C., Yu, L., Tong, H., He, C., **Wang, Y.**, and Fei, X.* (2026). Misbehaviour dominates GHG emissions from food loss and waste. **Nature Climate Change** (IF=30.3). https://doi.org/10.1038/s41558-026-02596-y (Citations: **3**)

CA-2 Ye, R., Yin, K., Yang, Z., Zhang, C., Huo, W., **Wang, Y.**, Zhang, H.*, Fei, X.*, 2026. A Novel Semi-Empirical Model for Multi-Scenario Estimations of Nitrous Oxide Emissions from Municipal Solid Waste Landfills. **Environmental Science & Technology** (IF=11.3), 60, 1, 595–608. https://doi.org/10.1021/acs.est.5c08871 (Citations: **1**)

CA-3 Yang, Z., Yin, K., Ye, R., Zhang, C., **Wang, Y.**, Zhang, H.*, Fei, X.*, 2025. Revealing hidden methane mitigation opportunities from legacy dumping and landfilling of municipal solid waste in rural China. **Journal of Environmental Management** (IF=8.4), 394, 127244. https://doi.org/10.1016/j.jenvman.2025.127244 (Citations: **4**)

CA-4 Yadav, V.*, Fei, X., Arora, M., van Emmerik, T.H.M., **Wang, Y.** and Laurent, A., 2025. Gaps in quantifying environmental losses of plastics impede effective solutions. **Nature Reviews Materials** (IF=79.8), pp.1-3. https://doi.org/10.1038/s41578-025-00802-5 (Citations: **6**)

CA-5 Wang, Y.J., Pi, X., Yadav, V., Hussain, A., Liu, Q., **Wang, Y.**, Guo, Y., Zhang, Y. and Fei, X.*, 2025. Leaching model of an expanding coastal dumpsite considering climate change. **Waste Management** (IF=8.1), 200, 114772. https://doi.org/10.1016/j.wasman.2025.114772 (Citations: **3**)

CA-6 Zhang, Z., Wang, J., Li, J., **Wang, Y.**, Yin, K. and Fei, X.*, 2024. Impacts of regional socioeconomic statuses and global events on solid waste research reflected in six waste-focused journals. **Waste Management** (IF=8.1), 182, pp.113-123. https://doi.org/10.1016/j.wasman.2024.04.028 (Citations: **7**)

CA-7 Pi, X., Fei, X.*, **Wang, Y.**, Sun, X. and Guo, Y., 2023. Global void ratio of municipal solid waste for compression indices estimation. **Waste Management** (IF=8.1), 160, pp.69-79. https://doi.org/10.1016/j.wasman.2023.02.003 (Citations: **12**)

CA-8 Lu, X., He, H., **Wang, Y.**, Guo, Y. and Fei, X.*, 2023. Masses and size distributions of mechanically fragmented microplastics from LDPE and EPS under simulated landfill conditions. **Journal of Hazardous Materials** (IF=13.6), 445, 130542. https://doi.org/10.1016/j.jhazmat.2022.130542 (Citations: **33**)

CA-9 Fei, X.*, He, H., Pi, X., Lu, X., Chen, Q., Ma, J., **Wang, Y.**, Fang, M., Wu, C. and Feng, S., 2023. The distribution, behavior, and release of macro- and micro-size plastic wastes in solid waste disposal sites. **Critical Reviews in Environmental Science and Technology** (IF=12.6), 53(3), pp.366-389. https://doi.org/10.1080/10643389.2022.2054649 (Citations: **57**)

CA-10 Fei, X.*, Guo, Y., **Wang, Y.**, Fang, M., Yin, K. and He, H., 2022. The long-term fates of land-disposed plastic waste. **Nature Reviews Earth & Environment** (IF=71.5), 3(11), pp.733-735. https://doi.org/10.1038/s43017-022-00354-0 (Citations: **63**)

CA-11 Fei, X.*, Fang, M. and **Wang, Y.**, 2021. Climate change affects land-disposed waste. **Nature Climate Change** (IF=30.3), 11(12), pp.1004-1005. https://doi.org/10.1038/s41558-021-01220-5 (Citations: **87**)

Note: * indicates the corresponding author. Full list of publications on:
ORCID: https://orcid.org/0000-0003-2373-0201
Google Scholar: https://scholar.google.com/citations?hl=en&user=aTPoRrMAAAAJ
Scopus: https://www.scopus.com/authid/detail.uri?authorId=57345240200.

Grants

- **Participant**, *Climate Resilience and Adaptation of Coastal Dumpsites and Landfills in East and Southeast Asia*. A*STAR–Chinese Academy of Sciences Joint Project (R25J3IR075), Feb 2026–present; approximately 200,000 USD; ongoing.
- **Participant**, *Mechanistic Modeling of Coastal and Riverine Solid Waste Disposal Sites in Southeast Asia and South Asia*. Singapore Ministry of Education Tier 1 Project (RG175/25), Feb 2026–present; approximately 170,000 USD; ongoing.
- **Participant**, *Environmental Standards and Engineering Behaviours of Reused Incineration Ashes from Municipal Solid Waste and Sewage Sludge*. Singapore National Environment Agency, Closing the Resource Loop Funding Initiative (CTRL-2025-1D-01), Mar 2025–present; approximately 4.8 million USD; ongoing.

- **Participant**, *"Breaking the Ground" in Semakau Landfill: Technological Solutions for Site Investigation and Material Reuse*. Singapore National Environment Agency, Closing the Resource Loop Funding Initiative (USS-IF-2021-4), Sep 2021–Sep 2024; approximately 4.7 million USD; completed.
- **Participant**, *Tackling Microfibres at Source: Investigating Opportunities to Reduce Microfibre Pollution from the Fashion Industry through Textile Design and Manufacturing Innovation*. United Nations Development Programme, Aug 2021–Aug 2023; approximately 45,000 USD; completed.

Media Coverage and Research Impact

Media coverage

- **NTU Engineering Annual Magazine 2024** selected my landfill methane research as one of five major research achievements featured in the College of Engineering's *Our Discoveries* section. The three-page feature, *Landfills are Trash for the Atmosphere* (pp. 12–14), highlights underestimated emissions and my improved estimation method accounting for waste composition, moisture, and temperature. [link]

Peer evaluation

- **Örjan Gustafsson** (Stockholm University), **Jiang Kejun** (Energy Research Institute, National Development and Reform Commission, China), and colleagues, in *Ten new insights in climate science 2024*, ***One Earth*** (2025), cited my research and recognized its contribution to integrating bottom-up and top-down approaches for methane emissions estimation. [link]
- **Daniel J. Jacob** (Harvard University) and colleagues, ***Environmental Research Letters*** (2025), cited my finding that rainfall alters waste decay rates to explain the satellite-observed surge in methane emissions at a US landfill during an abnormally wet year (2020). [link]
- **Joannes D. Maasakkers** (SRON) and **Daniel J. Varon** (MIT), with colleagues, ***Nature*** (2025), cited my research to demonstrate that landfill methane model parameters are uncertain and site-specific, supporting their case that independent satellite observations are critical for constraining landfill emissions. [link]
- **Tianhai Cheng** (Chinese Academy of Sciences) and colleagues, ***Nature Climate Change*** (2025), cited my research to support that meteorological variations drive fluctuations in landfill methane emissions, and that improved landfill management offers a more cost-effective mitigation pathway than equivalent reductions in the energy sector. [link]
- **Guanghao Chen** (Hong Kong University of Science and Technology) and colleagues, ***Nature Cities*** (2025), cited my research to establish that landfills generate more than 10% of global anthropogenic methane emissions, evidencing the environmental burden of conventional food-waste landfilling and helping make the case for rethinking separate food-waste and wastewater management in 29 large cities. [link]
- **Beidou Xi** (Chinese Research Academy of Environmental Sciences) and colleagues, ***Water Research*** (2026), cited my research to establish the substantial global climate contribution of landfill methane, helping highlight the climate stakes of micro(nano)plastic-driven disruption of carbon–nitrogen cycling in landfills. [link]

Policy and practical applications

- **Climate TRACE** adopted my improved landfill methane estimation method to determine site-specific waste decay rates. Its 2025 solid-waste methodology explicitly applies this approach in facility-level emissions estimation, translating the research into operational inventory calculations. [link]
- **WasteMAP / Rocky Mountain Institute (RMI)**: In collaboration with RMI, I contributed to the development of WasteMAP by integrating my improved landfill methane estimation method and mitigation assessment into the platform. I also supported data consolidation and landfill identification for city- and facility-level emissions accounting and decision support. [link]
- **OECD**, *Targeting methane emissions to mitigate the risk of climate overshoot* (2025), cited my research to evidence that landfill methane emissions are underestimated and mitigation opportunities are being missed, supporting its case for improved monitoring and cost-effective methane abatement. [link]

Service

Editorial role

- Young Editorial Board Member, *The Innovation Geoscience* [link]
- Young Editorial Board Member, *DeCarbon* [link]
- Young Editorial Board Member, *Exploration* [link]

Reviewer

- Nature Communications
- Journal of Environmental Management
- 9th International Congress on Environmental Geotechnics (ICEG)

Talk

- 2024 Frontiers of Environmental Science & Engineering (FESE) Young Talent Forum, Tsinghua University

Teaching

- CV0003 Introduction to Data Science and Artificial Intelligence, NTU, Y24/25, Semester 1 (52 hours)
  Lectures, Python laboratory sessions, quizzes and mini projects; student evaluation 8.0/10 (14 responses).
- CV1014 Introduction to Computational Thinking, NTU, Y24/25, Semester 2 (136 hours)
  Lectures, programming laboratory sessions and mini projects drawing on civil and environmental engineering.
- EG0001 Engineers & Society, NTU, Y23/24, Semester 1 & 2
  Facilitated discussions on engineering ethics, and supported assessment and feedback.

Student supervision

- Jia Linran, NTU, Environmental Engineering, PhD student (co-supervised with Assoc. Prof. Xunchang Fei): Reconciling bottom-up modelling with top-down monitoring of landfill methane emissions.
- Koh Xin Li, NTU, Environmental Engineering, Year 4, Final Year Project: Climate change effects on landfills.
- Steffi Rei Wong Shi Qi, NTU, Environmental Engineering, Year 4, Final Year Project: Carbon emissions from solid waste in Southeast Asia.
- Sim Zu Da Joel, NTU, Civil Engineering, Year 4, Final Year Project: Estimating landfill carbon stocks and emissions.
- Hou Tianyuan, NTU, Environmental Engineering, Year 4, Final Year Project: GHG emissions from waste sector.
- Soo Wei Song, NTU, Environmental Engineering, Year 4, Final Year Project: Method refinement for estimating carbon stock in landfills

Scholarships & Awards

- **Frontiers Planet Prize**, 2026
  The world's largest science prize in scientific solutions for planetary health. Sole first author of the selected *Nature Sustainability* paper: one of only 25 studies selected worldwide by a jury of 100 experts chaired by Professor Johan Rockström; Singapore's only selected study and the only one focused on solid waste globally.
- **Chinese Government Award for Outstanding Self-Financed Students Abroad**, 2025
  Highly selective China Scholarship Council award; 650 recipients worldwide.
- **Research Scholarship, Nanyang Technological University, Singapore**, 2022–2026
  Full tuition and monthly stipend.
- **Research Scholarship, Nanyang Technological University, Singapore**, 2020–2022
  Full tuition and monthly stipend.
- **NTU CEE Best Research Poster Award (Bronze)**, 2025
- **Tongji Scholarship of Excellence**, Tongji University, Academic Years 2016–2017, 2017–2018, 2018–2019